Cloud Computing: Architecture & Principles

Chapter 1: Definition and Advantages of Cloud Computing

Cloud computing is the on-demand delivery of IT resources (compute power, storage, databases, networking) via the Internet with pay-as-you-go pricing. According to NIST, essential cloud characteristics include on-demand self-service, broad network access, resource pooling, rapid elasticity, and measured service.

Key advantages of cloud computing include: - Cost Efficiency: Eliminates large capital expenditures (CapEx) for physical datacenters and hardware in favor of operational expenses (OpEx). - Elastic Scalability: Resources can dynamically scale up or down instantaneously in response to traffic spikes. - High Availability & Reliability: Redundant geographic zones provide fault tolerance and business continuity. - Rapid Global Deployment: Applications can be provisioned in multiple regions around the world within minutes.

Chapter 2: Cloud Service Models (SPI Model)

Comparing IaaS, PaaS, and SaaS

Cloud services are categorized into three primary layers of abstraction: 1. Infrastructure as a Service (IaaS): Provides fundamental computing resources including virtual machines, storage, and software-defined networks. The customer manages the operating system, middleware, and applications (e.g., AWS EC2, Google Compute Engine, Azure VMs). 2. Platform as a Service (PaaS): Provides hardware and application development software platforms hosted by the cloud provider. Developers can focus on code deployment without managing underlying infrastructure (e.g., AWS Elastic Beanstalk, Heroku, Google App Engine). 3. Software as a Service (SaaS): Delivers complete, ready-to-use software applications accessed over web browsers. The vendor handles all maintenance, infrastructure, and runtime (e.g., Google Workspace, Microsoft 365, Salesforce).